Chinyere E. Ugwuanyi

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EDUCATION

Rutgers University – Master of Business & Science (MBS) | GPA: 3.89 | *May 2026*

- **Concentration: Analytics & Data Science**
- **Relevant Courses**: Regression, BI & Visual Analytics, Data Warehousing, Data Management, Data Structures (Java)

West Virginia State University – B.S. Chemistry | GPA: 3.84 | *April 2021*

TECHNICAL SKILLS

- **Programming & Tools:** Python (Pandas, NumPy, Scikit-learn), R, SQL/MySQL, SPSS, Qualtrics, Tableau, Excel
- **ML & Statistics:** Regression, Classification, Survival Analysis, Reliability Testing, Experimental Design, Longitudinal and Cohort Analysis
- **Data Engineering & Validation:** Data Cleaning, Feature Engineering, Model Validation, Quality Assurance
- **Survey & Research Operations**: Multi-campus Survey Administration, Response Rate Optimization, Reliability Analysis, Incentive Protocol Design, Knowledge Transfer Documentation
- **Web Technologies**: HTML/CSS

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | *Rutgers - Office of Institutional Research & Decision Support (OIRDS)*
March 2025 – Present

- Managed end-to-end survey lifecycle across 3 campuses including Qualtrics setup, multi-wave distribution, response rate monitoring, and gift card lottery protocols for **40,000+** students.
- Identified and resolved a Qualtrics data quality issue affecting 950 respondents, maintaining data integrity across analysis workflows.
- Integrated survey responses with 941 university club records via institutional API, building fuzzy-matching and deduplication pipelines in R to standardize unstructured inputs.
- Performed large-scale roster QA using R, Python, SPSS, and Excel to ensure cross-campus data integrity.
- Built Tableau dashboards with multi-dimensional filters (college, major, degree level, campus) to support institutional decision-making.
- Conducted longitudinal cohort analyses on engagement, retention, and equity gaps to inform institutional planning.
- Developed an M-index dining utilization algorithm (n = **4,800+**, $p < 0.001$) supporting evidence-based resource allocation; research featured in Rutgers University news.

Student Researcher - Computer Science Education | *Rutgers Women in Computer Science*
Spring 2024 - Fall 2024

- Designed Qualtrics surveys (n = **900+**) to assess Computer Science and IT&I curriculum effectiveness.
- Built Tableau dashboards and visual reports supporting curriculum improvement recommendations.

Treasurer & At-Large Member | RUSA Allocations Board - Rutgers University
Fall 2024

- Supported equitable distribution of $1M+ funds to 400+ student clubs.
- Tracked budgets, allocation requests, and bi-weekly appeals processes.

PROJECTS

Rutgers University Student Organizations Survey Data Integration & Analysis | *Rutgers OIRDS* *(In Progress)*

- Integrated survey responses with official university club records by scraping institutional API data (941 organizations) and merging with respondent-provided text entries.
- Developed data-cleaning and fuzzy-matching pipelines in R to standardize club names, resolve duplicates, and link unstructured survey inputs to official organization identifiers.
- Produced a clean, analysis-ready dataset to support survey reporting, incentives distribution, and student engagement initiatives across multiple campuses.

ZIP-Code-Level COVID-19 Early Detection & Surveillance Model (NJ) *(In Progress)*

- Engineer dynamic transmission indicators (velocity, acceleration, jerk, rolling variance, volatility ratios, correlation) from daily ZIP-code- and county-level COVID-19 case data using left-sided and two-sided calculus-based derivatives to capture real-time epidemic dynamics.
- Characterize epidemic wave structure by identifying inflection points and critical points in 7-day moving average curves; validate using Wilcoxon rank-sum tests comparing velocity and acceleration distributions before and after candidate dates across 21 NJ counties.
- Apply multi-method drift detection (KS test, CUSUM, Isolation Forest ensemble) to identify structural regime shifts and emerging outbreaks earlier than traditional county-level models; benchmark detection lag across waves and counties to evaluate early warning performance against established epidemiological approaches to improve targeted intervention strategies.

WiDS Datathon 2026 - Wildfire Survival Modeling Project GitHub Repo

- Build discrete-time survival model predicting when wildfires will reach within 5km of evacuation zones; (risk horizons: 12h, 24h, 48h, 72h) to support emergency-response logistics.
- Expand events to person-period data, engineer time encodings, train gradient-boosted hazard model, evaluate with cross-validated Brier scores.
- Produce calibrated, time-specific risk curves to inform evacuation timing, resource staging, and operational planning under uncertainty.

Fake News Categorization – ML Pipeline (~45,000 articles) | *Accenture* Live Demo

- Built a text classification pipeline using TF-IDF, cosine similarity, and Nearest Neighbors to detect misinformation and reduce training bias and data leakage.
- Engineered 20+ linguistic features (e.g., punctuation density, sentence structure, readability scores) supporting interpretable modeling.

Airbnb NYC Price Prediction Model – Applied ML GitHub Repo

- Built and validated regression model (40K+ listings, 5-fold CV), identifying key drivers of NYC price variability.

SurveyList Webpage Redesign | *Rutgers OIRDS* Rutgers OIRDS Survey Research Program

- Rebuilt the OIRDS Survey Research Program webpage from scratch using HTML/CSS.
- Redesigned the architecture into audience-segmented sub-pages (undergraduate, graduate, faculty, staff) with structured data tables, navigation, and accessible layout; now live at oirds.rutgers.edu.

CERTIFICATIONS / PRESENTATIONS / ADDITIONAL INFORMATION

- **American Educational Research Association (AERA) Annual Meeting** (2026) – Co-author
  From Survey to Success: Predicting First-Year Retention Through Longitudinal Data Analysis
- **Rutgers University – Office of Institutional Research & Decision Support** - New Brunswick, NJ | *December 2025*
  Meet the Cohort of 2025: What Multi-Year New Student Survey Data Reveals About Our Newest Scarlet Knights
- **Cornell Tech** | **Break Through Tech AI/ML Program** - Machine Learning Foundations | *July 2025*
- **CITI Program** | **Social / Behavioral / Epidemiologic Research Investigators** | *June 2024*
- **LinkedIn Learning** – Business Analysis Foundations: Business Process Modeling, Data Analytics for Business Professionals, Turning Data into Strategies | *January 2026*